

Day 20

Even and Odd Parity Checker in Verilog

Verilog Code

A) Even parity checker

```
1 module even_parity_check(in,p,error );
2   input[3:0] in;
3   input p;
4   output error;
5   assign error=in[3]^in[2]^in[1]^in[0]^p;
6 endmodule
```

B) Odd parity checker

```
1 module odd_parity_check(in,p,error );
2   input[3:0] in;
3   input p;
4   output error;
5   assign error=~(in[3]^in[2]^in[1]^in[0]^p);
6 endmodule
```

Testbench Code

A) Even parity checker tb

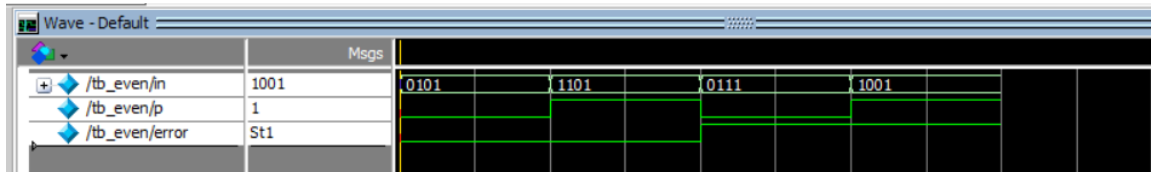
```
1 module tb_even;
2   reg[3:0] in;
3   reg p;
4   wire error;
5   even_parity_check dut(in,p,error);
6   initial
7   begin
8     in=4'b0101; p=1'b0; #10
9     in=4'b1101; p=1'b1; #10
10    in=4'b0111; p=1'b0; #10
11    in=4'b1001; p=1'b1; #10
12    $finish;
13  end
14 endmodule
```

B)Odd parity checker tb

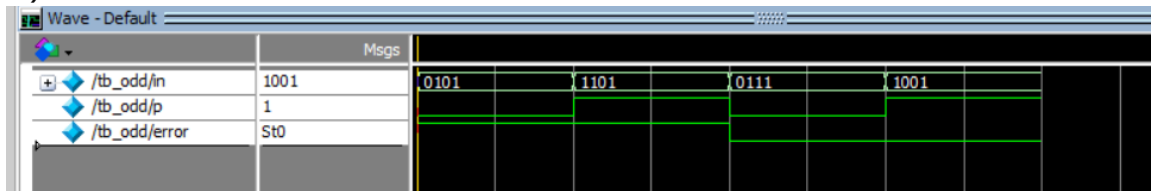
```
1 module tb_odd;
2   reg[3:0] in;
3   reg p;
4   wire error;
5   odd_parity_check dut(in,p,error);
6   initial
7   begin
8     in=4'b0101; p=1'b0; #10
9     in=4'b1101; p=1'b1; #10
10    in=4'b0111; p=1'b0; #10
11    in=4'b1001; p=1'b1; #10
12    $finish;
13  end
14 endmodule
```

Waveform

A)

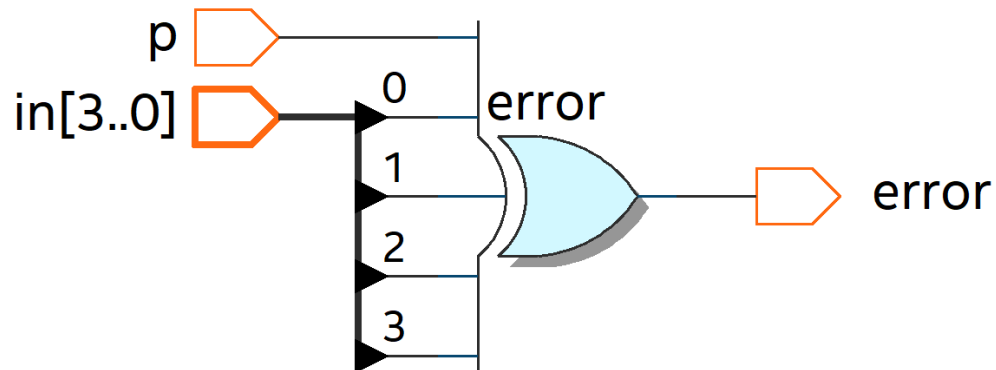


B)

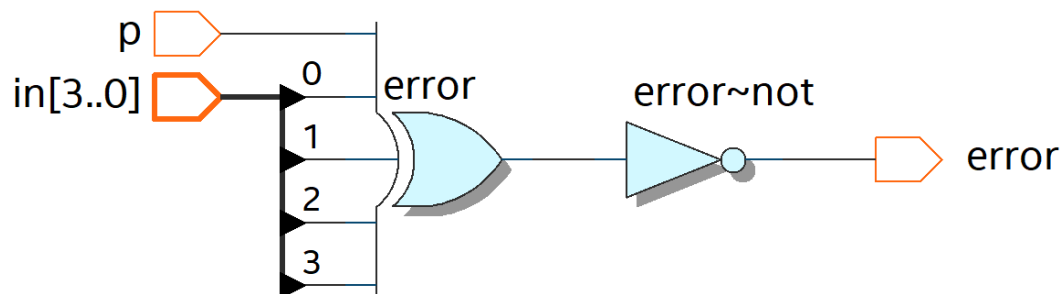


Schematic

A)



B)



EDA Tools Used

- IntelQuartusPrime
- ModelSim